MILITARY SPECIFICATION

VANADIUM PENTOXIDE
(FOR USE IN AMMUNITION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of Vanadium Pentoxide (V_2O_5) for use in lithium reserve type batteries (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipping and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

FSC: 6810

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E1 - Wire-Cloth Sieves For Testing Purposes
ASTM-E203 - Water, Using Karl Fischer Reagent
ASTM-E300 - Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Material.-The material shall be vanadium pentoxide (V_2O_5) conforming to the following requirements:

3.1.1 Form.-The material shall be in the form of yellow-brown crystalline powder and shall be free of dirt, chips, grease and other foreign matter when examined as specified in 4.4.3.1.

3.1.2 X-ray diffraction.-The material shall conform with the X-ray diffraction pattern for V_2O_5 as given in the ASTM Diffraction Data File when determined as specified in 4.4.3.2.

3.1.3 Chemical Properties.-The material shall conform with the following chemical properties:

Property	Requirement	Applicable Paragraph
Moisture Content,%,max	0.10	4.4.3.3
Vanadium (as V_2O_5),% min(on Moisture Free Basis)	99.90	4.4.3.4
Chlorides (as Cl),%max	0.007	4.4.3.5
Iron, (as Fe_2O_3),% max	0.015	4.4.3.6
Sodium (as Na_2O),% max	0.025	4.4.3.7
Silica (as SiO_2),% max	0.088	4.4.3.8

3.1.4 Particle Size.-The particle szie of the material shall conform to the following when determined in accordance with 4.4.3.9.

Tyler Mesh Size:	60	100	150	200
%, max, Retained on :	0.0	25.0	60.0	85.0
%, min, Retained on :	0.0	0.0	25.0	50.0

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3.2 First article testing.-This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 2 pounds of V_2O_5 obtained by sampling as described in 4.4.2.1. The sample shall be obtained from a production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4.3 of this specification.

4.3.3 Rejection.-See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Lot formation.-A lot shall consist of one or more batches of V_2O_5 produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of V_2O_5 that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105.

4.4.2 Testing

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4.4.2.1 Sampling.-Approximately 1 pound of V_2O_5 shall be withdrawn from each batch to be sampled using ASTM Procedure E300-70 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed should be tested in accordance with the applicable methods given in paragraph 4.4.3. If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The samples shall be subjected to the tests listed below:

Inspection/Requirement	Defect Classification
Form (see 3.1.1)	Major B
X-ray Diffraction (see 3.1.2)	Major B
Properties (see 3.1.3)	Major B
Moisture	
Vanadium (as V_2O_5)	
Chlorides (as Cl)	
Iron (as Fe_2O_3)	
Sodium (as Na_2O)	
Silica (as SiO_2)	
Particle Size (see 3.1.4)	Major B

4.4.2.2 Inspection Equipment.-The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicalbe for all tests and examinations specified in 4.4.3. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of inspection equipment is required prior to its use for acceptance purpose.

4.4.3 Test methods and procedures.-(see 6.2) The tests in 4.4.3.2 thru 4.4.3.5 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.4.3.1 Form.-Spread approximately 1/2 pound of the sample on a flat surface and examine the material visually for the presence of contamination and form.

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4.4.3.2 X-ray Diffraction.-An X-ray diffraction pattern of the material shall be obtained using a powder X-ray diffraction technique described in a standard or reference textbook on X-ray diffraction.

4.4.3.3 Moisture Content.-Determine the moisture content of the sample using the Karl Fischer method as described in ASTM E203.

4.4.3.4 Vanadium

4.4.3.4.1 Solutions

1:1 H_2SO_4
Ortho-Phosphoric Acid
Ferrous Ammonium Sulfate 0.1 Normal (approximately)
Potassium Permanganate 0.1 Normal (approximately)
Urea (C.P. salt)
Barium Diphenylamine Sulfonate - 0.3 g dissolved in 100 ml of hot water. Decant the clear solution.
Sodium Nitrite - 3.45 g per liter
Potassium Dichromate - Government Standard #136B

4.4.3.4.2 Standardization of the Ferrous Ammonium Sulfate.-Weigh 0.1250 g of Potassium Dichromate and transfer to a 600 ml beaker. Add 250 ml of H_2O, 40 ml of 1:1 H_2SO_4 and stir until dissolved. Add 10 ml of phosphoric acid and cool to about 60°F. Add 10 drops of barium diphenylamine sulfonate and titrate with 0.1 normal ferrous ammonium sulfate to the first green color. (Color change is sharp from a purple to blue to green).

$$\text{Calculation: Relation Factor} = \frac{25.492}{\text{Ml of Ferrous Ammonium Sulfate}}$$

4.4.3.4.3 Procedure.-Transfer 0.4500 g (on a Moisture Free Basis) of sample to a 600 ml beaker. Add 40 ml of 1:1 H_2SO_4 and slowly heat until dissolved. Evaporate to light fumes. Cool, add 100 ml of H_2O bring to a boil, add 10 ml of phosphoric acid, 150 ml of H_2O and cool to about 60°F. in a water bath. Add 0.1N potassium permanganate until a definite pink color is obtained which persists for about 2 minutes. Add sodium nitrite solution from a buret until the pink color is destroyed, then add 1 ml excess. Immediately add 2 g of urea and stir. Add 1 ml of barium diphenylamine sulfonate and titrate with 0.1 N ferrous ammonium sulfate until the first green end point is obtained. (Color change from purple to blue to green). Record the ml of titrant used.

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4.4.3.4.4 Calculation

$\% V_2O_5 = A \times B \times 2.021$

Where: A = Ml of ferrous ammonium sulfate.
B = Relation factor

4.4.3.5 Chlorides.-Determine the chloride content in accordance with a turbidimetric method described in a standard or reference textbook (see 6.4).

4.4.3.6 Iron.-Determine the iron content as follows:

4.4.3.6.1 Solutions:

.2% 1.10 Orthophenanthroline Monohydrate
10% Hydroxylamine Hydrochloride
1 Molar Sodium Perchlorate
Sodium Sulfate (anhydrous)
Nitrobenzene - Purifed Grade
Standard Fe solution - 1 ml = 10 micro grams. Dissolve 0.0702 g pure $Fe(NH_4)_2(SO_4)_2 \cdot 6\ H_2O$ in 100 ml of 5% HCl and dilute to 1000 ml.

4.4.3.6.2 Procedure.-Transfer 0.250g of the sample into a 100 ml beaker. Add 10 ml of HCl and heat slowly until sample is dissolved. Add 10 ml of H_2O, cool to room temperature, add 20 ml. of hydroxylamine hydrochloride and continue as in preparation of calibration curve. Since hydroxylamine hydrochloride forms a strong complex with Vanadium (IV) which at a pH of 4-5 gives a colorless solution, the use of pH paper is not necessary in the pH adjustment of the sample solution. The correct pH is attained by adding ammonia until the solution is just colorless. A blank should be run along with the sample. Care should also be taken in the selection of glassware. It is often wise to set aside glassware and reagents to be used for this determination of iron only.

4.4.3.6.3 Preparation of Calibration Curve.-Transfer 0 (Blank), 1, 2, 4. 6, 8 and 10 ml of the standard iron solution to 100 ml beakers. Dilute to 10 ml with water, add 20 ml of hydroxylamine hydrochloride and let stand for about 8 minutes. Add 1-1 ammonium hydroxide from a burette until a pH of 4-5 is reached. (pH paper can be used). Quantitatively transfer the solution to a 250 ml

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separatory funnel, add 1 ml of sodium perchlorate, 20 ml of ortho-phenanthroline and shake. Add exactly 30 ml of nitrobenzene and shake vigorously for 30 seconds. Let the phases separate and draw off the nitrobenzene layer into a clean 50 ml beaker containing 1g of sodium sulfate. Shake and immediately filter through a small Whatman #54 paper into a photometer cubette. Immediately cap the cubette and read the absorbance on a photometer at 525 mu, using the blank to zero the photometer. Proceed as above on all the calibration solutions and plot the readings against micrograms of iron. The V.C.A uses a spec 20 photometer or equivalent.

4.4.3.7 Sodium.-Determine the sodium content in the sample in accordance with an Atomic Absorption method described in a standard or reference textbook. Convert the percent sodium found to Na_2O.

4.4.3.8 Silica.-Determine the silica content of the sample as follows:

Transfer a 5.00 gram sample into a 600 ml beaker. Add 80 ml of 1-1 H_2SO_4 and 20 ml $HClO_4$. Cover the heat slowly until the sample is dissolved then evaporate until dense fumes of perchloric acid escape freely from the spout of the beaker.

Cool, dilute to about 450 ml with water, and filter through a Whatman #40 filter paper containing a small wad of filter paper. Wash the paper and precipitate alternately with hot 1:4 HCl and hot water about 6 times. Transfer the paper and precipitate into a platinum crucible and ignite in an electric muffle furnace, first at low heat and then at a higher temperature to obtain constant weight. Cool, weigh the crucible, then add 1 ml of water, 3 drops of sulfuric acid and 2 ml of hydrofluoric acid. Place the crucible on a hot plate and slowly evaporate to dryness, then heat in the blast of a burner. Reweigh the crucible, and record the loss in weight.

$$\%SiO_2 = \frac{\text{Loss in Weight}}{5} \times 100$$

4.4.3.9 Particle Size.-Superimpose the sieves of the meshes required in 3.1.4 as defined in ASTM E11 and attach the receiving pan to the bottom. Place 100 grams of the sample in the top sieve, mount the assembly on a shaker and operate the shaker for 5 minutes. At the end of 5 minutes weigh the granules retained in each sieve and in the pan and calculate the percentages of each.

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A and Level C.-Unless otherwise specified, the V_2O_5 shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first detination for immediate use.

5.2 Marking.-Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data.-See MIL-A-48078 (PA)

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ATTN: SARPA-QA-A-P, Picatinny Arsenal, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Vanadium Pentoxide which has been found satisfactory for use in lithium batteries is manufactured by Foote Mineral Company, Exton, Pa. 19341. V_2O_5 from other sources should be evaluated by Picatinny Arsenal, Dover, New Jersey 07801 (ATTN: SARPA-QA-A-M) before being approved as an equivalent.

6.4 Method available from Hellige, Inc., 877 Stewart Ave, Garden City, New York.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 6810-AA71

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